PHYS 525 Project Report

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1.1 Introduction

Some of the topics we have covered in this course have been the derivation of the Larmor radius and its effect on $E \times B$ drift, the treatment of plasmas as fluids, and the process of diffusion of these fluids characterized by diffusion equations derived from the fluid models of plasmas. In nuclear fusion research one current problem is gradient driven turbulence, where random microscopic oscillations perpendicular to a background temperature or density gradient induce an $E \times B$ drift that amplifies the oscillation. This turbulence is highly chaotic and frequently generates sudden spikes in the plasma's local density and temperature. Over the course of this project we decided to study how an isolated, turbulence-induced spike in temperature or density diffuses through the HSX stellarator in its Quasi-Helically Symmetric (QHS) mode (it should be said that we planned to do a comparison between the way temperature or density spikes diffuse in HSX's QHS configuration and its mirror configuration however we could not get the .mgrid magnetic mesh files for the mirror configuration because Dr. Boeyaert was travelling). Specifically, we were first curious how a density spike would defuse along a field line so we wrote a 1D diffusion equation solver to solve a diffusion equation over the parallel connection length of the HSX stellarator since we knew that the spike would diffuse quite quickly and so would defuse over a shorter length than the parallel connection length. We calculated the parallel connection length using Dr. Heinke Frerichs's FLARE and MOOSE code bases and HSX .mgrid and boundary files from Dr. Dieter Boeyeart and we observed the way varying the plasma temperature changed the spike decay time. We then extended the scope of the project to observing how heat diffuses across magnetic field lines. To do this, we modified our diffusion equation solver to use an FTCS scheme in cylindrical coordinates instead of cartesian coordinates as in the density diffusion equation solver and we derived a different heat flow diffusion equation using a similar method as we used to get the diffusion parameter for the density diffusion equation. We then used FLARE and MOOSE to calculate the magnetic field strength at every point in a set of different cross sections across one half of HSX so we could observe the way the varying magnetic field in different sections of HSX affected the diffusion of heat away from the core of the plasma. In this study we varied both the toroidal angle at which we calculated the magnetic field strength across the whole cross section at that toroidal angle and the plasma temperature for each toroidal angle.

1.2 Methods

Diffusion Equations

The fundamental logic of the general diffusion equation is based on two core physical principles. The first is Fick's First Law which states that the flux of particles into a differential volume element of plasma will be proportional to the difference in the density between two adjacent volume elements. The second principle is that the instantaneous rate of change of the density at any position within the plasma will be the spatial derivative of the flux at that point because the difference in the flux leaving the volume element and entering the volume element at that position gives the total density change in the volume element. In other words, the time derivative of the density at a particular point in space is equal to the space derivative of the flux and the flux is the space derivative of the density.

For our specific density diffusion equation, because we curious to see how quickly a density spike diffused in a plasma in HSX, we assumed a weakly ionized plasma that has reached steady state where the plasma is isothermal and so derived our diffusion equation parameter in the exact same way the textbook in chapter 5 does it. Intuitively, the result we got for our diffusion parameter equation makes sense because the rate of diffusion will be less if the collisions with neutral atoms happen more frequently because that slows down the movement of the electrons and the diffusion rate will be greater if the starting velocity of the electrons is greater to begin with so the product of the speed of the electrons and the time per collision gives you something along the lines of the distance electrons travel before they collide and the greater that distance, the greater the diffusion rate. In other words, because the thermal velocity effects both the space derivative of density and the space derivative of flux, the diffusion parameter is proportional to the thermal velocity squared and because just the flux is inversely proportional to the collision frequency, the diffusion parameter is inversely proportional to the collision frequency.

For our specific heat flow diffusion equation, we made many of the same assumptions as we did for the density diffusion equation (plasma is weakly ionized, plasma is steady state, plasma is isothermal) except now since the heat was flowing across field lines in a collisionary manner we knew the magnetic field would affect the diffusion parameter. So, we derived our diffusion parameter in the exact same manner that the textbook did in section 5.5 and then we assumed the plasma was operating such that $\omega_c^2\tau^2 \gg 1$ such that our diffusion parameter became $\frac{KT\nu}{m\omega_c^2}$ and then we rearranged that expression using the definition of the cyclotron frequency and the Larmor radius so the mechanism of the diffusion inhibition would be clear in the equations. One very important assumption that we made is that the magnetic field exhibited radial symmetry so that we could model the diffusion with a 1D diffusion equation instead of a 2D diffusion equation.

<u>Finding Parallel Connection Lengths</u>

The parallel connection length is the distance a particle would travel if it were to start at the wall of a reactor and follow one of the magnetic field lines of the reactor's magnetic field until it collided with the wall of the reactor. In order to calculate the parallel connection length we used Dr. Heinke Frerichs's FLARE function named fieldline_connection. This function accepts an array of starting coordinates and integrates the path along the local magnetic field vector in both the forward and backward directions. As it does this it determines the distance travelled in either direction along the magnetic field line until it comes into contact with a prespecified boundary. It then sums the distance travelled in either direction to determine the parallel connection length. We specified the magnetic field within HSX and the boundary of HSX using .mgrid and boundary files. We also specified the starting positions using numpy arrays, initializing 50 points at the midplane of the plasma in the scrape-off layer. The radii values of the points were chosen to identify the parallel connection lengths of the scrape-off layer which is a region containing open magnetic field lines outside the confined plasma core plasma which rapidly collide into the metal vacuum vessel.

<u>Finding Magnetic Field Strength</u>

In order to calculate the magnetic field strength at specific positions within HSX we used Dr. Heinke Frerichs's FLARE function named bfield.eval. This function takes an array of spatial coordinates and performs a calculation with the prespecified magnetic field values in the .mgrid file. This file contains the magnetic field values at a number of coordinates and the eval function uses these values to perform a 3d spatial interpolation to approximate the magnetic field at the positions we provided. We chose specific toroidal angles and generated a numpy array of positions at the midplane of the plasma and then provided these points to the eval function to determine the magnetic field strength at these positions.

<u>Finite Difference Methods</u>

For both our diffusion equation solvers we used Forward Time Center Space (FTCS) finite difference methods however for our first diffusion equation solver (density diffusion equation) we used a FTCS scheme in cartesian coordinates whereas for our heat flow diffusion equation solver we used an FTCS scheme in cylindrical coordinates. We used insulated boundary conditions in both cases however for the density diffusion equation solver we derived a boundary condition based on the idea that we solved the equation over an entire parallel connection length and so since the stellarator is closed, the flux leaving one "end" of the field line had to be entering the other "end" of the field line. In other words, the flux at one end of the FTCS scheme was equal to the negation of the flux at the other end and we derived a boundary condition equation based on that logic using a forward divided difference to approximate the space derivative at the start of the array which we set equal to the negation of a backwards divided difference used to approximate the space derivative at the end of the array. It is also worth mentioning that we solved our diffusion equations in both studies assuming a gaussian spike in density/temperature and added the time and space varying fluctuating temperature or density to a baseline temperature or density set by the user which we chose for the density to be the operating plasma density of HSX and we varied the baseline temperature as part of our studies.

<u>Equations</u>

- $\frac{\partial \bar{n}}{\partial t} = \frac{\partial}{\partial s}(D_{||}(s)\frac{\partial \bar{n}}{\partial s})$
  - $\bar{n}$ is the average fluctuating component of the plasma density
  - s is the distance from an origin along a field line
  - $D_{||}(s) = \frac{v_{th,e}^{\ 2}}{\nu_{ei}}$ is the parallel diffusion coefficient as a function of distance along the field line
    - $v_{th,e} = \sqrt{\frac{T_e e}{m_e}}$ is the electron thermal velocity

        - $\nu_{ei} = \frac{n_e e^4 ln\Lambda}{3\pi^{3/2}\epsilon_0^{\ 2} m_e^{\ 1/2}(2T_e e)^{3/2}}$
        - NOTE: T_e is in eV
- $\frac{\partial \bar{T}}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}(r\chi\frac{\partial \bar{T}}{\partial r})$
    - $\bar{T}$ is the average fluctuating component of the plasma temperature
    - r is the radius from the core of the plasma to a particular point inside the cross section at a particular toroidal angle
    - $\chi = r_L^{\ 2}\nu_{ei} = \frac{m_e^{\ 2} v_{th}^{\ 2}}{e^2 B^2}\nu_{ei}$
        - $v_{th,\,e} = \sqrt{\frac{T_e e}{m_e}}$ is the electron thermal velocity
            - NOTE: T_e is in eV
        - $\nu_{ei} = \frac{n_e e^4 ln\Lambda}{3\pi^{3/2}\epsilon_0^{\ 2} m_e^{\ 1/2}(2T_e e)^{3/2}}$
            - $\Lambda = 12\pi n \lambda_D^{\ 3}$
            - $\lambda_D \equiv (\frac{\epsilon_0 K_b T_e}{ne^2})^{1/2}$ with $K_b T_e$ in joules
        - B is the magnetic field as a function of distance from the core of the plasma at a specific cross section inside HSX selected at a specific toroidal angle

Specific Study Details

For the density turbulence study, we used a parallel connection length of 1.33 meters and baseline plasma temperatures of 5,6,7,8,9, and 10 eV. The total time the plasma was simulated for was $5 * 10^{-8}$ seconds.

For the heat flow turbulence study, we calculated the magnetic field strength for cross sections with a toroidal angle of 0.0, 0.2, 0.4, 0.6, and 0.8 radians and for each of these angles, we used baseline plasma temperatures of 20, 40 ,60, 80 and 100 eV.

We set the number of spatial grid points to always be 1000 and the number of timesteps we used varied depending on the specific experiment but are documented in the tables below (the bracketed numbers). We always used a baseline plasma density of $2 * 10^{17}\ m^{-3}$. We set the threshold at which we considered the spike to have decayed to be the baseline temperature or density depending on the study plus $\frac{1}{e}$ times the max spike height.

1.3 Data

Figure 1: Graph of flare code output for parallel connection length of HSX stellerator at various start radius points. (NOTE: the region of the graph where the data is fixed at parallel connection lengths of 10000 m corresponds to the core of the plasma, where parallel connection lengths

approach infinity. We limited the simulation to 10000 m for computation purposes).

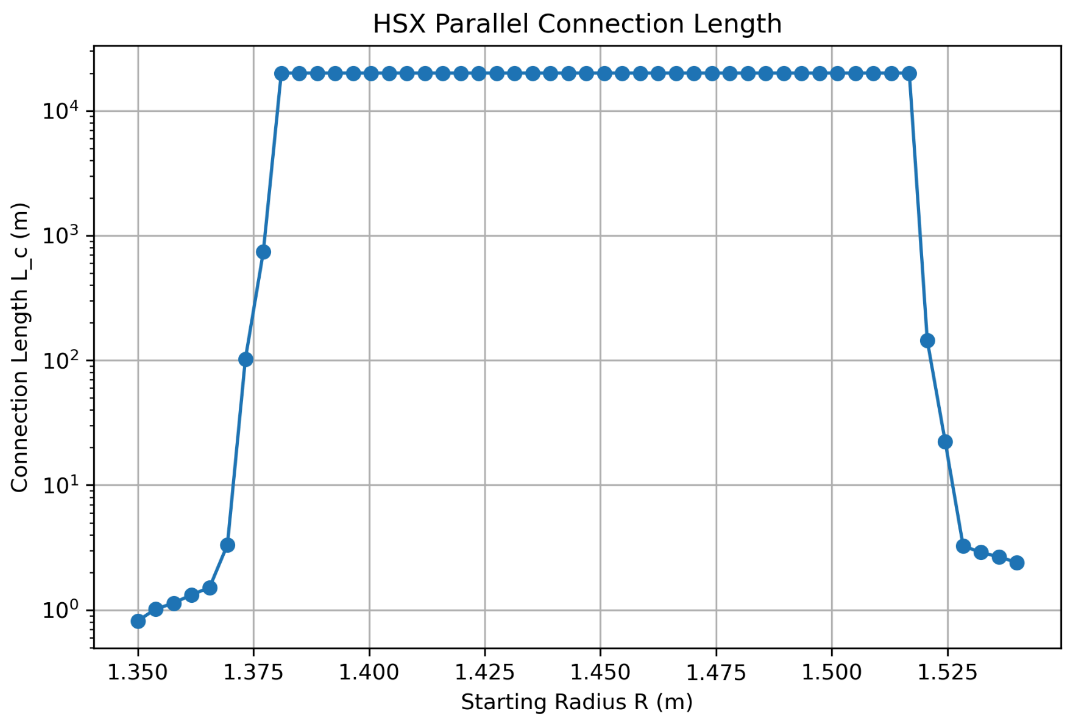


Table 1: Simulated diffusion of density turbulence spike at different plasma temperatures for a parallel connection length of 1.33 meters (simulation length). (NOTE: numbers in gray boxes are spike decay times in seconds and the number of timesteps used (numbers in brackets) for each simulation)

	Baseline Plasma Temperature					
Parallel Connection Length	5 eV	6 eV	7 eV	8 eV	9 eV	10 eV
1.33 m	8.5e-9 s (75000)	5.51e-9 s (115000)	3.81e-9 s (170000)	2.77e-9 s (228000)	2.09e-9 s (302000)	1.63e-9 s (390000)

Figure 2: Graph of density turbulence spike decay time at different plasma temperatures for a parallel connection length of 1.33 meters (simulation length).

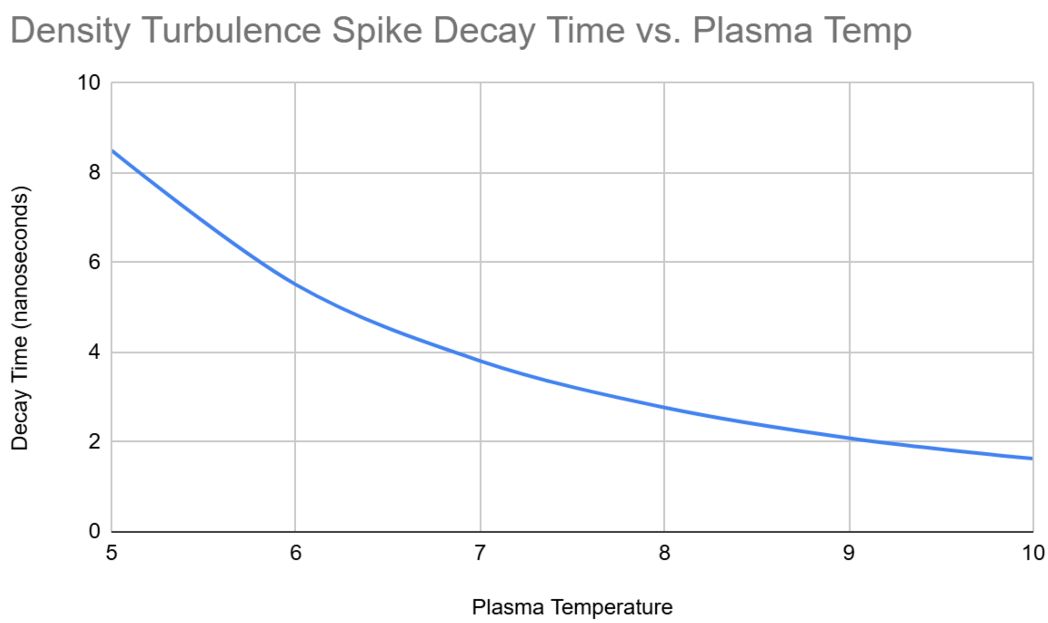


Figure 3: Graphs of HSX radial magnetic field strength profiles for different toroidal angles (0 rad - 0.8 rad)

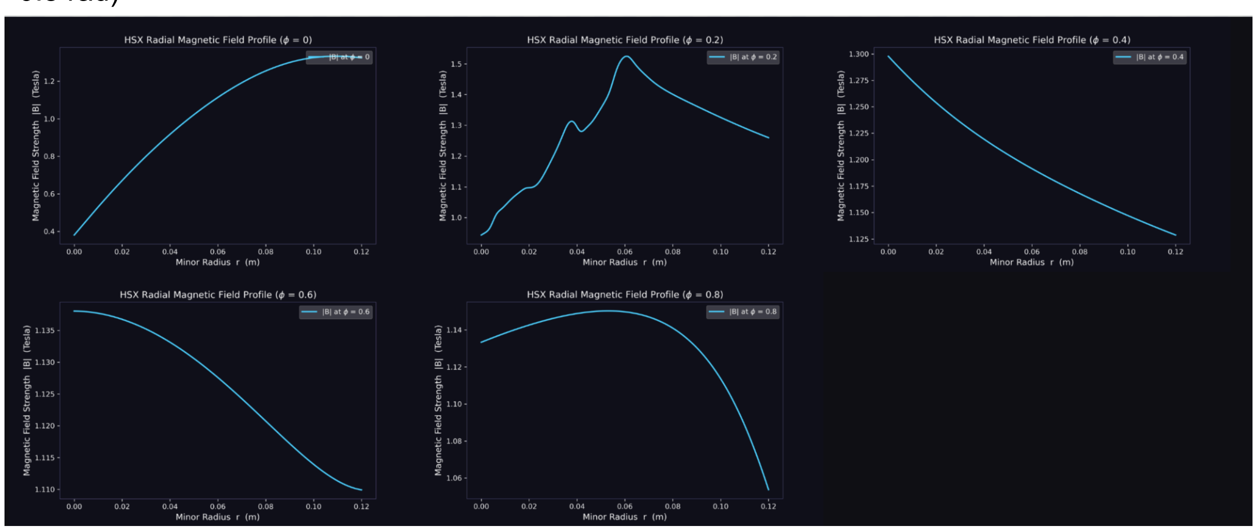
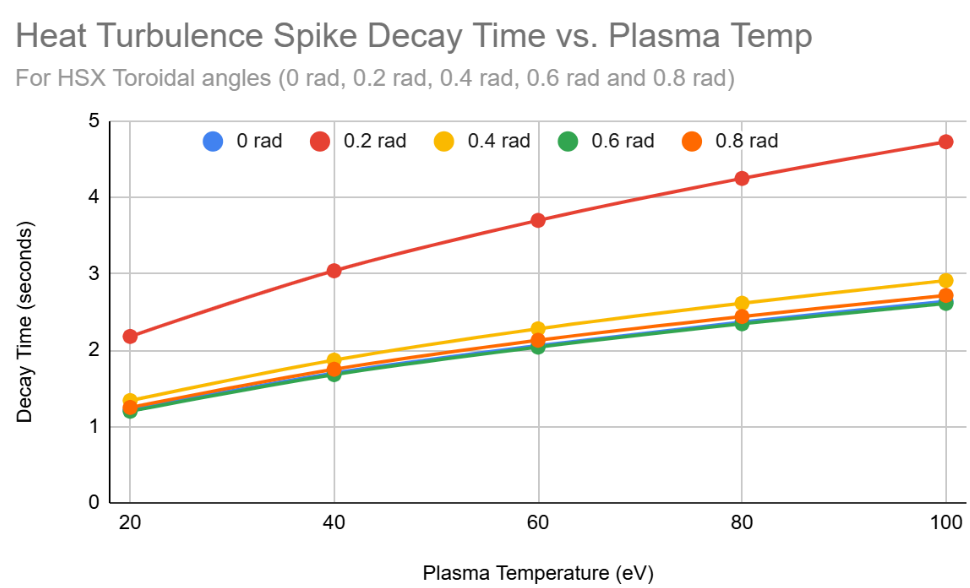


Table 2: Simulated diffusion of heat turbulence spike at different plasma temperatures (20 eV - 100 eV) for a set of toroidal angles (0 rad - 0.8 rad). (NOTE: numbers in gray boxes are spike decay times in seconds and the number of timesteps used (numbers in brackets) for each simulation)

		Baseline Plasma Temperature				
		20 eV	40 eV	60 eV	80 eV	100 eV
Toroidal angles	0 rad	1.22 (281000)	1.70 (201000)	2.06 (166000)	2.37 (145000)	2.64 (130000)
	0.2 rad	2.18 (47000)	3.04 (33000)	3.70 (28000)	4.25 (24000)	4.73 (22000)
	0.4 rad	1.34 (33000)	1.87 (24000)	2.28 (19000)	2.615 (17000)	2.912 (15000)
	0.6 rad	1.20 (34000)	1.68 (24000)	2.04 (20000)	2.346 (17100)	2.613 (16000)
	0.8 rad	1.25 (37000)	1.75 (27000)	2.13 (22000)	2.441 (19000)	2.718 (17100)

Figure 2: Graphed diffusion of heat turbulence spike at different plasma temperatures (20 eV - 100 eV) for a set of toroidal angles (0 rad - 0.8 rad). (NOTE: each line corresponds to a different toroidal angle that was analyzed for the HSX stellerator)



1.4 Analysis

For the density study, we observed in the data that the decay time decreased (meaning the disturbance propagated faster) as the plasma temperature increased. This makes sense because according to our diffusion equation, the diffusion parameter is proportional to the plasma temperature to the power of $5/2$ meaning the decay time should be approximately inversely proportional to the plasma temperature to the power of $5/2$ which lines up with what we observe.

For the heat study, we observed in the data that the decay time increased (meaning the disturbance propagated slower) as the plasma temperature increased. This makes sense because according to the diffusion equation for heat, the diffusion coefficient (χ) is proportional to the plasma temperature to the power of $-1/2$ meaning the decay time should be approximately proportional to the plasma temperature to the power of $1/2$. This is consistent with the observed data as when the plasma temperature increases the decay time increases as well. We also observed in the data that the decay time varied corresponding to which toroidal angle was chosen. This makes sense because unlike a tokamak, the HSX stellerator is non symmetrical and so at different toroidal angles within the stellarator the shape of the magnetic coils (and the corresponding magnetic field strength) is unique. This affects the heat diffusion by modifying the Larmor radius of electrons at different toroidal angles within the plasma. Since the diffusion coefficient (χ) is proportional to the square of the Larmor radius (and so inversely proportional to the square of the magnetic field strength (B)), at toroidal angles where the magnetic field is locally weaker the decay time is shortened and at toroidal angles where the field is locally stronger the decay time is lengthened.

Github Link

https://github.com/Lavansurendra/PHYS525_Project

References

Introduction to Plasma Physics and Controlled Fusion Third Edition, Francis Chen